

Victor Sorgi Pedroso

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EDUCATION

Georgia Institute of Technology

M.S. in Robotics (GPA: 3.75/4.0)

Atlanta, GA

Expected May 2027

Relevant Coursework: Advanced Mobile Robotics, Linear Control Systems, Kalman Filtering, Dynamics of Mechanical Systems

Emory University

B.S. in Physics and Astronomy (GPA: 3.61/4.0)

Atlanta, GA

May 2025

SKILLS

Programming: Python (NumPy, SciPy, Matplotlib, PyTorch, Astropy, CVXPY), C++, MATLAB

Controls & Robotics: LQR, PID control, Kalman filtering / EKF, convex optimization (LMIs), AirSim

Design & Tools: SolidWorks, Git, Excel VBA

Languages: Portuguese (Fluent), English (Fluent), Spanish (Advanced)

WORK EXPERIENCE

Volvo Autonomous Solutions

Autonomous Vehicles Intern

Greensboro, NC

May 2026 – Aug 2026

- Built a Python pipeline to find truck stops in MCAP logs and map where and how long they occurred. Recorded log names and timestamps so engineers could investigate each stop.
- Built a Claude API assistant that searches Confluence and Jira to answer engineering questions and cite its sources. The full team of 6–7 people adopted it.
- Compared U.S. route temperatures and estimated road grades with the trucks' tested limits to identify promising routes for autonomous trucking and flag those needing more investigation.
- Wrote a VBA macro to create partner release notes in Word from Excel, cutting hours of work to minutes per run.

Yamaha Motor Corporation

Intern → Associate Product Manager

Kennesaw, GA

May 2024 – May 2026

- Used Python and Excel VBA to convert technical specifications from Japan into U.S. formats and load them into the InRiver PIM database, reducing manual work and errors during annual product updates.
- Created a product-development timeline to clarify each team's responsibilities and make coordination easier.

RESEARCH EXPERIENCE

C3U Lab, Georgia Tech

Graduate Researcher

Atlanta, GA

Jan 2026 – Present

- Implemented a direct data-driven LQR controller in CVXPY and tested it on quadrotor dynamics in AirSim, designing persistently exciting inputs and checking closed-loop stability with LMIs.
- Planning a master's capstone on how autonomous vehicles can use simple messages to coordinate and avoid collisions.

Dr. Merida Batiste's Lab at Emory University

Undergraduate Researcher

Atlanta, GA

Jan 2022 – May 2025

- Used GALFIT for an honors thesis on how galaxy models affect effective-radius estimates and the $M - \sigma$ relation.

PROJECTS

Autonomous Mobile Robot Simulation

Independent Project

2026 – Present

- Built a Python simulation of a differential-drive robot with process noise and two PID controllers to guide it to a goal.
- Added an Extended Kalman Filter to estimate the robot's state from a motion model and noisy position measurements.